Forecasting tomorrow's temperature

I compared several forecasting models to see how well they predict temperature one day ahead.

Prepared by Aamir Sohaib

JENA WEATHER DATASET

Five days of history / Predict 24 hours ahead

BEST HELD-OUT TEST RESULT

2.317 °C mean absolute error

Across 1,730 unseen test examples, the 64-unit LSTM had an average error of about 2.32°C per forecast.

BASELINE (VALIDATION)	BEST (VALIDATION)	REDUCTION IN ERROR
2.586°C	2.416°C	6.6%
Same temperature tomorrow	LSTM with 64 units	Versus validation baseline

The recurrent models picked up useful weather patterns. Adding units or combining layer types, however, did not consistently improve accuracy.

The question and the data

I tested whether five days of weather data could predict the temperature 24 hours after the final observation.

420,551
original observations

70,129
hourly observations

14
weather measurements

How the weather data was prepared

01 I started with observations recorded every 10 minutes in Jena, Germany.

02 I averaged the readings by hour to make the sequences easier to compare.

03 I replaced 38 missing markers and filled gaps from preceding observations; 88 hours had no temperature readings.

04 I standardized the inputs using statistics from the training period only.

Time-ordered separation

I trained the models on older observations, selected a model using later data, and reserved the newest observations for an unseen-weather test.

SET	HOURS	START	END
Training	49,090	Jan 1, 2009	Aug 8, 2014
Validation	10,519	Aug 8, 2014	Oct 20, 2015
Testing	10,520	Oct 20, 2015	Jan 1, 2017

Each example uses 120 hourly readings (five days) to predict one temperature 24 hours later. I started a new example every six hours and kept its input and target in the same split.

Six approaches, one metric

I compared the models using validation mean absolute error (MAE), measured in °C. Lower values mean smaller errors.

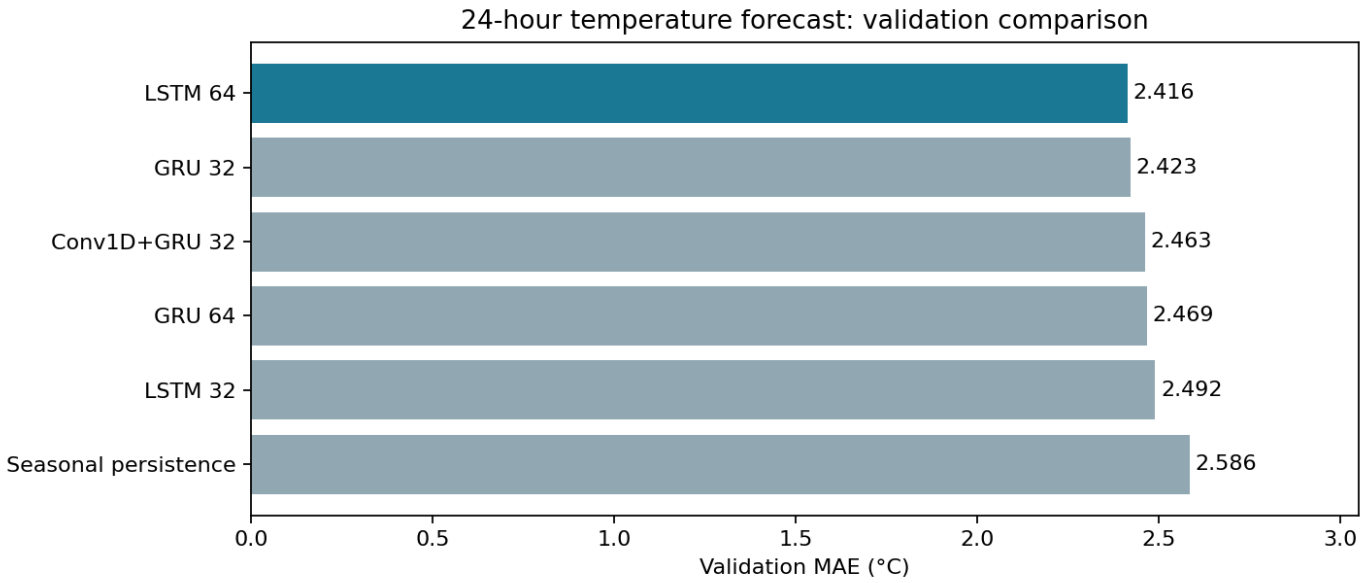


Figure 1. All five neural models had lower validation error than the seasonal-persistence baseline.

MODEL	VALIDATION MAE
LSTM 64	**2.416 °C**
GRU 32	2.423 °C
Conv1D+GRU 32	2.463 °C
GRU 64	2.469 °C
LSTM 32	2.492 °C
Seasonal persistence	2.586 °C

The 64-unit LSTM had the lowest validation MAE: 2.416°C versus 2.586°C for the baseline. That is a 6.6% reduction, or about 0.171°C less error on average.

What changed as the models learned

I trained each model for eight epochs. The figure traces its validation error during training.

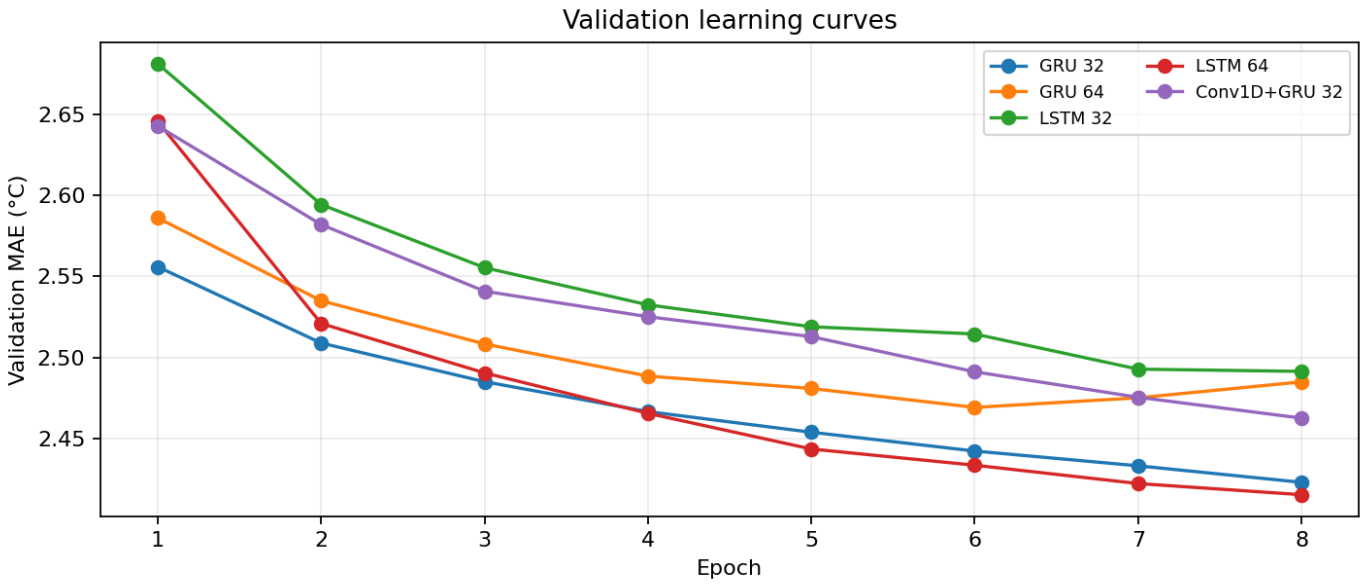


Figure 2. Validation error generally fell during training. All five models completed the eight recorded epochs.

DO MORE UNITS HELP?

Increasing the LSTM from 32 to 64 units reduced validation MAE from 2.492 to 2.416°C. For the GRU, the same change raised MAE from 2.423 to 2.469°C.

DO MORE LAYERS HELP?

The optional Conv1D + GRU model reached 2.463°C. It beat the baseline, but neither of the two leading recurrent models.

INTERPRETATION

Here, architecture mattered more than size alone. Some curves were still improving near epoch eight, so I compare the recorded runs rather than fully tuned versions of each model.

Testing the two leading models

I ranked the models on validation data, then tested the 64-unit LSTM and its close runner-up, GRU 32, on unseen weather.

LSTM 64 / TEST MAE

2.317°C

Best validation model

GRU 32 / TEST MAE

2.339°C

Validation runner-up

The 64-unit LSTM's test MAE was 0.022°C lower than GRU 32's (about 0.9%). That small gap in one run does not establish that LSTM always performs better.

SAMPLE TEST FORECAST

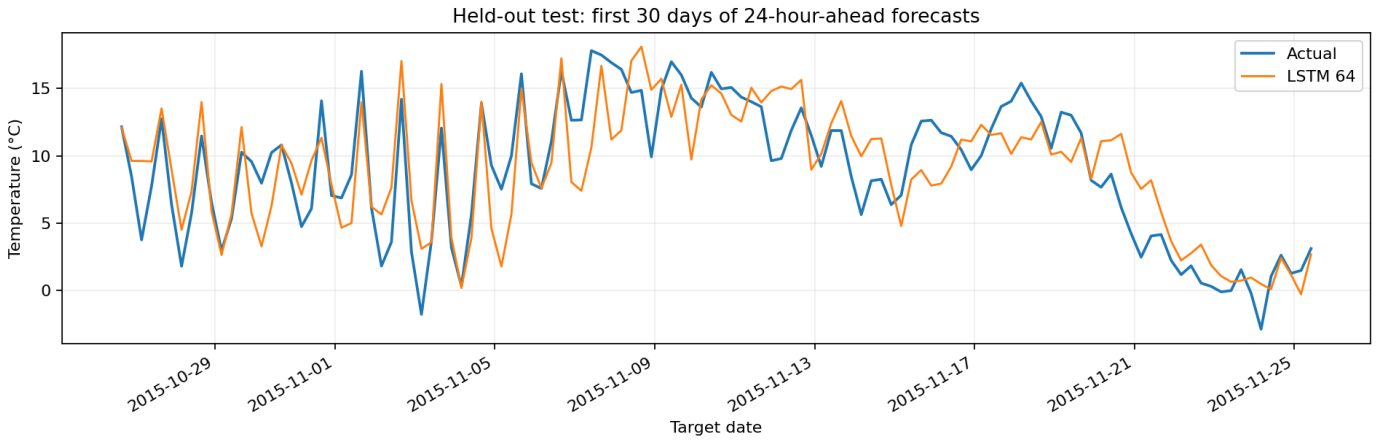


Figure 3. The first 30 days of held-out forecasts. Blue shows the observed temperature and orange shows the 64-unit LSTM prediction; this is only part of the test period.

HOW TO READ THE TEST SCORE

Across 1,730 unseen forecast windows, the LSTM's average error was 2.317°C. Individual forecasts can miss by more or less than that.

The chart plots only LSTM forecasts. The GRU 32 test score comes from the revised notebook, not from a GRU forecast file in the older export.

What the experiment tells us

I draw four conclusions from these forecasting experiments.

01 Forecasting beats the simple baseline

On validation data, all five neural models beat the simple rule of using the latest temperature as tomorrow's forecast. The leading LSTM reduced average error by 6.6%.

02 LSTM 64 was best in this run

The 64-unit LSTM had the lowest validation MAE (2.416°C) and a lower held-out test MAE than GRU 32 (2.317°C versus 2.339°C).

03 Larger is not always better

More units helped the LSTM but hurt the GRU in this comparison. A larger model did not automatically make better forecasts.

04 Extra architecture did not guarantee improvement

The Conv1D + GRU combination beat the baseline but fell short of the strongest single-recurrent-layer models.

SCOPE AND LIMITATIONS

This comparison uses one fixed-seed run, one chronological split, a six-hour forecast stride and eight training epochs. The LSTM-GRU gap is small; repeated runs or longer tuning could change the ranking. I did not perform those tests here.

The recurrent models learned useful patterns from recent weather and improved 24-hour forecasts. The 64-unit LSTM performed best in this experiment, but greater complexity alone did not predict better accuracy.